

MANJOT SRAN

Website ◇ GitHub ◇ LinkedIn ◇ Google Scholar

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RESEARCH INTERESTS

My research focuses on reliable multimodal AI for prediction across heterogeneous data sources. I develop and analyze fusion architectures, including early, intermediate, and late fusion; attention and gating mechanisms; mixture-of-experts models; and shared-private representation learning, with an emphasis on robustness to noise, missing modalities, and distribution shift. I further study learning under limited supervision and reliability-centered evaluation, integrating deep-learning-based fusion with uncertainty-aware soft-computing methods, particularly tolerance near sets, to characterize and manage ambiguity in multimodal classification and regression.

EDUCATION

University of Manitoba, Winnipeg, Canada

Sept. 2026 – Present

Ph.D. Student in Computer Science

Supervisors: Dr. Mengjun Hu and Dr. Sheela Ramanna

The University of Winnipeg, Winnipeg, Canada

Jan. 2024 – May 2026

M.S. in Applied Computer Science

GPA: **4.375/4.5**

Thesis: *SPriG-TLC: A Novel Framework for Shared-Only Fusion with Private Feature Regulation and Tolerance-Based Ambiguity-Aware Clustering for Multimodal Affective Computing*

Thesis Link: <https://winnspace.uwinnipeg.ca/handle/10680/2370>

Thesis passed with distinction

Supervisor: Dr. Sheela Ramanna, Professor, Applied Computer Science Department, The University of Winnipeg

Relevant Coursework:

- Advanced Machine Learning
- Web and Document Databases
- Pattern Recognition
- Digital Image Processing

IKG Punjab Technical University, Jalandhar, India (IKGPTU)

June 2019 – June 2023

B.Tech. in Computer Science and Engineering

Graduated with Distinction

CGPA: **8.83/10 (88.3%)**

Tier-1 accredited program by NBA (National Board of Accreditation, India)

Relevant Coursework:

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| • Data Structures | • Machine Learning |
| • Artificial Intelligence | • Database Management Systems |
| • Digital Electronics | • Wireless Sensor Networks |
| • Design and Analysis of Algorithms | • Computer Architecture and Microprocessor |

PUBLICATIONS

1. Manjot S. Sran, Sheela Ramanna
SPriG: Shared-Only Fusion with Improvement-Guided Private Gating for Multimodal Affective Computing. *Information Fusion* (Elsevier), published online July 7, 2026, Article 104606. DOI: 10.1016/j.inffus.2026.104606 [Impact Factor: 17.4].
2. Manjot S. Sran, Sheela Ramanna, Ketan Kotecha
DiMoE: Disentangled Representation Learning with Mixture-of-Experts Fusion for Sentiment Intensity Prediction and Emotion Classification. *Algorithms* (MDPI), 2026, 19(9), Article 743. DOI: 10.3390/a19090743 [Impact Factor: 2.6].
3. Manjot S. Sran, Sheela Ramanna
TLC-STRIDE: An Ambiguity-Aware Framework for Multimodal Sentiment Prediction. *Machine Learning with Applications* (Elsevier), [Minor Revision Under Review], 2026 [Impact Factor: 6.1].

AWARDS & HONORS

Graduate Studies Scholarship , The University of Winnipeg	<i>May 2026</i>
Graduate Student Research Award (GSRA) , The University of Winnipeg	<i>April 2026</i>
• Competitive \$7,000 graduate research award recognizing student-led research excellence.	
Graduate Student Knowledge Mobilization Grant (\$750)	<i>April 2026</i>
The University of Winnipeg	
Graduate Studies Scholarship , The University of Winnipeg	<i>Aug. 2025</i>
Graduate Studies Scholarship , The University of Winnipeg	<i>May 2025</i>
Graduate Studies Scholarship , The University of Winnipeg	<i>July 2024</i>

SKILLS

Programming Languages:	Python, C, C++, JavaScript, Solidity
Web Development:	HTML, CSS, Bootstrap, Django, Flask, jQuery, XML
Networking:	Cisco Packet Tracer
Data Analysis:	Excel, Spreadsheets, SQL, NumPy, Pandas, Matplotlib
Machine Learning and Deep Learning:	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, torchvision, librosa
LLM/GenAI Frameworks:	LangChain, LangGraph, LlamaIndex, RAG
Systems:	Linux, Git, Docker
Communication Skills:	Strong written and verbal communication skills

RELEVANT EXPERIENCE

Guest Instructor, Computer Science	<i>Sept. 2026 – Present</i>
Department of Computer Science, University of Manitoba - Winnipeg, Manitoba, Canada	
• Serve as a Guest Instructor for selected lectures in COMP 3360/DATA 3010 and COMP 4710 during the Fall 2026 term. • Prepare and deliver assigned course lectures in the absence of Prof. Mengjun Hu, covering scheduled material in machine learning and data mining.	
Teaching Assistant, Data Science with Real World Data Sets (DATA 3010 B01)	<i>Sept. 2026 – Present</i>
Department of Computer Science, University of Manitoba - Winnipeg, Manitoba, Canada	
• Support instruction for DATA 3010 during the Fall 2026 term, with a focus on applied data science using real-world datasets. • Facilitate lab sessions and guide students through practical data science workflows, dataset processing, and analysis techniques.	

- Assist with coursework evaluation and grading, and hold office hours to provide individualized student support.

Teaching Assistant, Applications of Database Systems (ACS-2814-001) *Sept. 2025 – Dec. 2025*

University of Winnipeg - Winnipeg, Manitoba, Canada

- Led weekly labs and tutorials on ER modeling, normalization, and SQL (DDL/DML) with query design and optimization.
- Provided one-on-one support during office hours and troubleshoot student issues with tools such as Access and SQL Server.
- Assisted with assessment administration (quizzes/exams) and delivered actionable feedback to improve learning outcomes.

Participant (Pathfinder Cohort)

June 2025 – Sept. 2025

Winnipeg, Manitoba, Canada

- Participated in North Forge's Founders Program (Pathfinder cohort) to accelerate an early-stage digital health venture.
- Conducted user research with caregivers of people living with dementia to identify unmet needs, workflow gaps, and support preferences.
- Collaborated with mentors and peer founders to refine business model, value proposition, and go-to-market approach.
- Prepared prototype demos and delivered progress updates and pitches to North Forge stakeholders.

Senior Research Assistant

Aug. 2024 – Nov. 2024

University of Winnipeg - Winnipeg, Manitoba, Canada

- Collaborated on the preparation and submission of an NSERC grant application, ensuring alignment with funding agency guidelines.
- Drafted and revised key proposal sections, including research objectives, methodologies, and projected impact.
- Managed deadlines and provided administrative and editorial support to finalize the application on time.
- Ensured high-quality content through iterative reviews and close communication with the lead researcher.

ACADEMIC & PROFESSIONAL SERVICE

Webmaster, International Joint Conference on Rough Sets (IJCRS 2027)

Sept. 2026 – Present

University of Manitoba, Winnipeg, Manitoba, Canada

- Develop and maintain the IJCRS 2027 conference website in coordination with the organizing committee.
- Publish conference announcements, deadlines, program information, and organizational content, ensuring that information remains accurate and current.

RELEVANT INDUSTRIAL TRAINING EXPERIENCE

Guru Nanak Dev Engineering College, Ludhiana, India

July 2021 – Aug. 2021

Four Weeks of Industrial Training on Web Technologies

- Gained practical experience in web development technologies such as HTML, CSS, and JavaScript.
- Learned web development best practices, including responsive design, accessibility, and web security.

- Developed a website as part of the training program using HTML, CSS, JavaScript, and Bootstrap.

National Institute of Electronics & Information Technology (NIELIT), Chandigarh, India

July 2022 – Aug. 2022

Six Weeks of Industrial Training on Blockchain Technology

- Learned about blockchain technology and its practical applications.
- Developed smart contract creation skills using Solidity.
- Studied decentralized application development and the Ethereum platform.
- Worked in a team to develop a blockchain-based project as part of the training program.

TCIL-IT, Chandigarh, India

Jan. 2023 – June 2023

Six Months of Industrial Training on Data Science and Machine Learning

- Gained practical skills in Data Science and Machine Learning.
- Explored and implemented machine learning libraries and algorithms.
- Learned how to fine-tune algorithm parameters and evaluate performance on real-world datasets.
- Completed a machine learning project on cricket score prediction as part of the training program.

RELEVANT PROJECTS

Project: Prompt-to-Video Generation Platform (SDXL + SVD with LLM Planning)

- Built a prompt-only AI video generation pipeline: user prompt → LLM-based scene planning → SDXL keyframe synthesis → Stable Video Diffusion rendering.
- Added reliability mechanisms including quality-gating and automatic fallback retries (seed and tuning adjustments) to improve coherence and reduce identity drift.
- Deployed an end-to-end service with FastAPI backend and Gradio UI for interactive video generation and result inspection.

Project: Feature Disentanglement + Mixture-of-Experts Fusion for Affective Computing

- Designed a shared-private disentanglement framework for multimodal affective computing.
- Implemented sparse top- k Mixture-of-Experts fusion with joint-router and separate-router variants.
- Evaluated on five benchmarks for sentiment intensity prediction and emotion classification across tri-modal and bi-modal settings.
- Performed ablation and interpretability analyses, including expert-usage and shared-versus-private contribution studies.

Project: Exploring Audio-Visual Fusion: Comparative Analysis of Fusion Strategies for Multi-Modal Emotion Recognition

- Explored audio-visual fusion using audio and video modalities.
- Conducted a comparative analysis of Early, Mid, and Late fusion strategies using Wav2Vec2 and VideoMAE.
- Achieved 95.39% accuracy on RAVDESS and 86.84% accuracy on CREMA-D with Late Fusion techniques.
- Demonstrated the strength of decision-level fusion approaches in multimodal tasks.

Project: KD-tree Construction and Search Algorithm Development

- Implemented KD-tree construction for managing k -dimensional data attributes.
- Developed Exact, Range, and Nearest Neighbor search algorithms.
- Optimized for speed and accuracy, comparing performance against linear search.
- Demonstrated significant efficiency gains with KD-trees over brute-force methods.

Project: Employee Management Systems Web Application

- Built with Python and Django framework.
- Implemented CRUD operations with Django ORM on SQLite.
- Designed a user-friendly UI for adding, deleting, filtering, and updating employee data.
- Ensured seamless data management and accessibility across the system.

RESEARCH SEMINARS & COLLOQUIA

Physics Colloquium, University of Winnipeg (organized by the Department of Physics)

Sept. 26, 2025

Winnipeg, Canada

Research Talk: Robust Alzheimer's Disease Detection via Multimodal Fusion

- Presented a multimodal AD diagnosis framework combining MRI (DeiT) with structured clinical variables (FT-Transformer), integrating modalities via early and mid fusion.

COMPETITIONS

North Forge RampUp 2025 — Pitch Competition

April, 2025

- Pitched a startup idea in front of 300+ attendees, engaging judges and mentors.

3MT (Three-Minute Thesis) Competition

March, 2025

- Presented “Advancing Emotion Recognition with Artificial Intelligence” at UWinnipeg’s annual Three-Minute Thesis competition.

VOLUNTEERING

Volunteer, UWinnipeg STEM Days

May 7–8, 2025

- Contributed to the STEM Days event organized by the University of Winnipeg.
- Delivered an outreach talk for Grade 7 students on May 7, 2025, presenting on Artificial Intelligence.

RELEVANT CERTIFICATIONS

- **Explainable Machine Learning (XAI)**, offered by Duke University *March 2025*
- **NLP: Twitter Sentiment Analysis**, certificate earned upon completion of a guided project offered by Coursera Project Network *March 2025*
- **Google AI Essentials**, offered by Google *Nov. 2024*
- **Deep Learning**, offered by Illinois Institute of Technology *Nov. 2024*
- **Critical Thinking for Better Decisions in the ChatGPT Era**, offered by Deep Teaching Solutions *Nov. 2024*
- **Data Science Math Skills**, offered by Duke University *Aug. 2024*
- **Deep Learning Applications for Computer Vision**, offered by University of Colorado Boulder *Aug. 2024*
- **Mathematics for Machine Learning: Multivariate Calculus**, offered by Imperial College London *Aug. 2024*
- **CS50 SQL Certificate**, offered by Harvard University *Dec. 2023*
- **CS50B Certificate**, offered by Harvard University *Sept. 2023*
- **CS50x Certificate**, offered by Harvard University *Aug. 2023*

- **Google Data Analytics**, offered by Google on Coursera *May 2023*
- **Python for Data Science and Machine Learning**, offered by Udemy *April 2023*
- **CS50's Intro. to Programming with Python**, offered by Harvard University *Feb. 2023*
- **Blockchain Technology Training Certificate**, offered by NIELIT, Chandigarh *Dec. 2022*

REFERENCES

Dr. Sheela Ramanna

Professor

Department of Applied Computer Science
University of Winnipeg, Winnipeg, Canada
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Dr. Yangjun Chen

Professor

Department of Applied Computer Science
University of Winnipeg, Winnipeg, Canada
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